



Md Ishmam Tasin

ishmam.tasin@bracu.ac.bd | +8801790226520

[linkedin](#) | [Official website](#) | [github](#)

Research Interest

Usable Security & Privacy; Human-Computer Interaction (HCI), Usable Security; AR/VR-based authentication; Cloud Computing for secure, scalable systems; Blockchain and Smart Contracts; Decentralized Systems; ICTD; socio-technical systems and technology adoption.

Educational Qualifications

B.Sc. in Computer Science

BRAC University

Dhaka, Bangladesh

Jan. 2019 – Sept. 2023

CGPA: 3.71 /4.00; Final 6 Semesters CGPA: **3.81/4.00**

- Thesis: Revolutionizing microfinance: a blockchain-driven decentralized finance (DeFi) model for collateral-free loans [[Institutional Repository](#)], Thesis Advisor: [Professor Md Sadek Ferdous](#)
- Relevant Courses: Cryptography and Crypto Analysis, Blockchain and Cryptocurrency, Cloud Computing, Programming for the Internet, System Analysis and Design, Database System Design, Microeconomic Theory, Advanced Anthropology, Business Communications

Language Proficiency

- **English:** TOEFL iBT Score: 99/120 (Reading: 25/30, Listening: 25/30, Speaking: 24/30, Writing: 25/30) [[PDF](#)]

Professional Experiences

Research Intern at Penn State University (Remote)

May. 2026 – Present

College of Information Science & Technology, Pennsylvania State University

- **Lab:** Human Signal in Intelligence, Verification, and Empowerment (HIVE) Lab
- **Research Works:**
 - * Currently leading a research on age verification technology and non-expert behavior and attitude around different age verification methods and how that affect their comprehension of privacy and security perception.

Graduate Research Assistant (Full Time)

Oct. 2023 – Present

Dept. of CSE, BRAC University

- **Supervisor:** [Dr. Farida Chowdhury](#), Professor, Department of Computer Science and Engineering, BRAC University.
- **Teaching Experience:**
 - * Mentored undergraduate thesis students by explaining research processes and methodologies, defining scope and theoretical frameworks, conducting regular progress meetings, and assisting in planning future research directions.
 - * Assisted my supervisor in conducting and evaluating lab sessions for the Technical Communication (CSE391) course, guiding students through the research process including LaTeX, literature reviews, and writing research papers across domains such as HCI, security and privacy, socio-technical aspects, blockchain, gamification.
 - * Conducted seminar and learning session on research methodology as part of [Human Centered Computing Society \(HCCS\)](#) Research Group at BRAC University.

- * Prepared research-related resources and tutorials for students interested in topics including HCI, blockchain, AR/VR, and usable security and privacy.

• Research Works:

- * Conducted research on blockchain-based decentralized finance (DeFi) and self-custodial crypto wallets, focusing on security, usability, and inclusivity for marginalized and low-literate users.
- * Conducted qualitative (interviews, focus group discussions, contextual inquiries) and quantitative (surveys) studies on **Blockchain, Crypto Wallets**, socio-technical issues involving women, marginalized communities, low-literate users, people with disabilities and accessibility needs, and users of advanced technologies such as cryptocurrencies.
- * Conducted user studies on proposed and existing solutions to evaluate effectiveness, efficiency, acceptance, reliability, adoption, comfort, and trust, and to identify user challenges and preferences.
- * Assisted my supervisor in preparing **Research Grant Proposal** (over 150k USD) for the Improving Computer and Software Engineering Tertiary Education Project (ICSETEP), funded by the University Grants Commission of Bangladesh.
- * Developed and evaluated XR-based authentication systems (integrating VR/AR) to mitigate shoulder-surfing and keystroke inference attacks, validated through formal analysis and user studies (n=30).
- * Held regular meetings with my supervisor to discuss ongoing research progress and brainstorm future directions in HCI, usable security and privacy in different domains, and XR-based solutions.
- * Mentored undergraduate thesis students on publishing papers in peer-reviewed conferences and journals by analyzing research scope, theoretical contributions, and engagement with existing literature.
- * **Research Area:** Blockchain, HCI, ICTD, AR/VR, Usable Privacy and Security, Cloud Computing.
- * **Technology Stack:** JavaScript, TypeScript, Solidity, Node.js, Express.js, Next.js, MySQL, MongoDB, Git, Docker, Smart Contracts, Ganache, Truffle, Mocha, Chai, , EVM, EIP, Non-custodial Wallet, Draw.io, Miro, JIRA

Software Developer Intern

Oct. 2021– Feb. 2022

Friendship NGO

- Contributed to the design and development of mHealth applications for underprivileged rural communities, conducting qualitative interviews with users, NGO officials and analyzing findings to derive design requirements and functionalities for effective digital interventions.
- **Technology Stack:** JavaScript, TypeScript, React.js, Angular.js, Git, Power BI, Figma, GCP, UserTesting, SurveyMonkey, and Google Form

Research & Publications

Accepted Papers:

- [1] **AuthVR: Securing Authentication Against Shoulder Surfing and Keystroke Inference Attacks using Virtual Reality** [PDF] [Link]
 - **Author:** Md. Yeasin Ali, Touhid Rahaman Uday, Md Ishmam Tasin, Masum Alam Nahid, Fairuz Rahaman Chowdhury, Farida Chowdhury, Md Sadek Ferdous.
 - **Conference:** Published in 24th IEEE International Conference on Trust, Security and Privacy in Computing and Communications ([TrustCom 2025](#))
- [2] **A Secure Augmented Reality (AR) Based Authentication Mechanism For ATM** [PDF] [Link]
 - **Author:** Touhid Islam Uday, Md Yeasin Ali, Md Ishmam Tasin, Masum Alam Nahid, Fairuz Rahaman Chowdhury, Hossain Shahria, Farida Chowdhury, S M Taiabul Haque, Md Sadek Ferdous.
 - **Journal:** Published in [IEEE Access](#).
- [3] **Self-Custody in an Unsupported Ecosystem: Usability and Security Challenges Among Novice and Experienced Crypto Wallet Users in Bangladesh** [PDF]
 - **Author:** Md Ishmam Tasin, Md Sadek Ferdous, S M Taiabul Haque, Wolfgang Prinz, Farida Chowdhury.
 - **Conference:** Accepted at International Conference on Computer-Human Interaction Research and Applications ([CHIRA 2026](#))
- [4] **Why Can't I Just Use a Username? Understanding Why Users Struggle with Messaging DApps** [PDF]

- **Author:** Tithi Saha*, Md Ishmam Tasin* (* denotes equal contribution), Ahad Mahjabin Jannat, Md Masum, Farida Chowdhury.
 - **Conference:** Accepted The 8th IEEE International Conference on Decentralized Applications and Infrastructures ([IEEE DAPPS 2026](#))
- [5] **Navigating the Cybersecurity Maze: Women’s Practices, Awareness and Vulnerabilities in Bangladesh** [\[PDF\]](#)
- **Author:** Maliha Tasnim, Maisha Tasnim, Samsad Laila, Seham Al Haque, Tanzuma Tabassum, Md Ishmam Tasin, S M Taiabul Haque, Farida Chowdhury.
 - **Conference:** To appear in 9th International Conference on Computer-Human Interaction Research and Applications ([CHIRA 2025](#))
- [6] **From Agent Dependency to Financial Autonomy: Smartphone-Based Inclusive MFS Design for Low-Literate Population** [\[PDF\]](#)
- **Author:** Yousuf Abdullah Harun, Asiful Islam Chowdhury, Nurshida Akter Nilima, Hafsa Bintey Alim, Ashfiqu Ahmed Miftah, Md Ishmam Tasin, S M Taiabul Haque, and Farida Chowdhury.
 - **Conference:** To appear in Conference on Computing and Sustainable Societies ([COMPASS 2026](#))
- [7] **A Proposed Systematic Framework and Guideline of Birth Declaration in Rural Last-Mile Bangladesh: Aimed at Reducing Child Marriage** [\[Link\]](#) [\[PDF\]](#)
- **Author:** Shabab Intishar Rahman, Priom Deb, Md Ishmam Tasin, Md. Sadiqul Islam Sakif, Jannatun Noor.
 - **Conference:** Proceedings of the 7th ACM SIGCAS/SIGCHI Conference on Computing and Sustainable Societies ([COMPASS 2024](#))

Under Review/ Ongoing Work:

- [1] **“Not Every SOS Means Call the Police”: Survivor Agency and Trauma-Informed Design Mandate** [\[PDF\]](#)
- **Author:** Atika Hossain Ainun*, Md Ishmam Tasin* (* denotes equal contribution), Rifah Tasnim Mahin, Lamiya Islam, Jawaril Munshad Abedin, S M Taiabul Haque, Farida Chowdhury.
 - **Conference:** Under review (2nd round) in ACM CHI conference on Human Factors in Computing Systems ([CHI 2027](#))
- [2] **TrustChain: A Decentralized Protocol for Inclusive Microfinance Using Trust Circles and Social Collateral** [\[PDF\]](#)
- **Author:** Md Ishmam Tasin, Md. Rabib Hossain, Nahin Chowdhury, S.M.Azwad-Ul-Alam, Md Sadek Ferdous.
 - **Conference:** Ongoing work to be submitted to Advances in Financial Technologies ([AFT 2026](#))
- [3] **“Why Johnny Still Doesn’t Use Encryption in Bangladesh”: A Usability Perspective** [\[PDF\]](#)
- **Author:** Mashia Hossain Mim, Fahad Bin Mahbub, Md Ishmam Tasin, Md Sadek Ferdous, Md Forhad Rabbi, Farida Chowdhury.
 - **Conference:** Under review in The 21st ACM ASIA Conference on Computer and Communications Security ([ACM ASIACCS 2026](#))
- [4] **The Exiled Prince: An Octalysis-Guided Visual Novel Game for Promoting Cybersecurity Awareness among Non-Technical University Students** [\[PDF\]](#)
- **Author:** MD Atiq Mahbub, Mirza Md. Shafi Uddin, Nouruzzaman Niloy, Md Ishmam Tasin, Md. Tawsif Shahriar, Md Sadek Ferdous, Jannatun Noor, Farida Chowdhury
 - **Conference:** Under review (2nd round) in 11th IEEE European Symposium on Security and Privacy ([Euro S&P 2026](#))
- [5] **Designing for Rationality: Behavioral Economics Design Paradigm for Safer E-commerce Browsing Mandate** [\[PDF\]](#)
- **Author:** Md. Samsul Arefin, Jannatul Ferdous Nawrin, Irfanul Hoque, Istiak Islam, Mohammad Jawadul Tashick, Md Ishmam Tasin, Md Sadek Ferdous, S M Taiabul Haque, Farida Chowdhury.
 - **Conference:** Under review in ACM Designing Interactive Systems ([DIS 2026](#))

- [6] **Towards Usable Cybercrime Reporting Systems Through the Usability Principles: A Study in the Context of Bangladesh** [\[PDF\]](#)
- **Author:** Taskia Raushan Ferdousi, Omar Shahariar, Tasfia Shobnom Anika, Md. Ibna Sohayab, Nowshin Nusrat Neha, Md Ishmam Tasin, Jannatun Noor Mukta, and Farida Chowdhury.
 - **Conference:** In Conference on Computing and Sustainable Societies ([COMPASS 2026](#))
- [7] **Trust, Recognition, and Agency in Tech-Mediated Peer-to-Peer Communication within the Global South** [\[PDF\]](#)
- **Author:** Md Ishmam Tasin, Mehedi Hassan Alif, Sherajum Mysha, Md. Mehedi Hasan Shishir, Md. Sabbir Ahmed, and Farida Chowdhury.
 - **Status:** To be submitted to the ACM CHI Conference on Human Factors in Computing Systems ([CHI 2027](#))
- [8] **Usability, Privacy, and Trust in Large Language Models: Perspectives from Non-Technical Professionals in Academia, Banking, and Healthcare** [\[PDF\]](#)
- **Author:** Md Ashraful Hoque, Ahsanul Fahim Ahmed, Fahim Ahmad, Nuzhat Zahedi Promity, Md Ishmam Tasin, Md. Sabbir Ahmed, and Farida Chowdhury.
 - **Status:** To be submitted to the ACM CHI Conference on Human Factors in Computing Systems ([CHI 2027](#))
- [9] **“When Satire Becomes Generative”: Understanding Satirical Mediation in the Age of AI in the Global South**
- **Author:** Anonymous.
 - **Conference:** Ongoing work, planned for submission to the ACM CHI Conference on Human Factors in Computing Systems ([CHI 2027](#)).
- [10] **“Beyond the Bait”: Gamified Interventions for Phishing and SMiShing Awareness in Bangladesh’s Evolving Cyber Ecosystem**
- **Author:** Anonymous.
 - **Conference:** Ongoing work, planned for submission to the ACM CHI Conference on Human Factors in Computing Systems ([CHI 2027](#)).

Academic Services

Peer Reviewing

- **ACM CHI 2027:** Serving as a peer reviewer for research papers submitted to the ACM CHI Conference on Human Factors in Computing Systems.
- **ACM COMPASS 2026:** Reviewed 2 research papers submitted to ACM SIGCAS/SIGCHI Conference on Computing and Sustainable Societies - COMPASS 2026.
- **IEEE BCCA 2026:** Reviewed 3 research papers submitted to International Conference on Blockchain Computing and Applications.
- **HCCS Lab, BRAC University:** Serving as an Executive Member of the Human-Centered Computing and Society (HCCS) Lab.

Awards & Recognitions

Dean’s Recognition List	2019-2023
• Honourable mention in the Dean’s recognition list for excellent performance as undergraduate student for 6 semesters.	
Graduated with High Distinction	2023
• Awarded to candidates for academic excellence achieved over the duration of an entire undergraduate career.	
Runner Up: Affine Blockchain Hackathon 2022	2022
• BdOSN and Affine had partnered with Wind and Anchorblock, two leading Web3 companies from Bangladesh to host this competitive Hackathon.	
Champion: DataBird Launchpad 2021	2021

- DataBird LaunchPad 2021 brought in exciting partnerships with ICT Division, Bangladesh, LICT Bangladesh, Durbar, BASIS, and The Daily Star, giving budding entrepreneurs and their innovative ideas the opportunity to shine on both international and local stages.

Finalist: Bangladesh Blockchain Olympiad (BCOLBD) 2021 2021

- The Blockchain Olympiad Bangladesh (BCOLBD) invited university students and recent graduates to submit blockchain solutions to solve real-world problems. BCOLBD 2021 was held from 25 to 27 February 2021.

Merit Scholarship in Undergraduate at Brac University 2019

- Awarded each semester to undergraduate students demonstrating outstanding academic performance.

National Merit Scholarship in Higher Secondary Certificate Exam (HSC) 2017

- A merit-based scholarship award granted by the Bangladesh Government to students who achieve exceptional performance on the Higher Secondary Certificate (HSC) examinations.

Golden A+ in both Secondary (SSC) and Higher Secondary Certificate Exam (HSC) 2015, 2017

- In Bangladesh, “Golden GPA” (or Golden A+) is a term used to describe achieving the highest grade, A+ (80-100 marks), in every single subject in the public examinations.

Online Certifications

- **Docker & Kubernetes - The Practical Guide** (Udemy, **Instructor:** Maximilian Schwarzmüller, 23.5 hrs, Sep 2021) [\[Link\]](#)
- **Master Ethereum & Solidity Programming** (Udemy, **Instructor:** Crystal Mind Academy, 9 hrs, Sep 2021) [\[Link\]](#)
- **The Git & GitHub Bootcamp** (Udemy, **Instructor:** Colt Steele, 17 hrs, Sep 2021) [\[Link\]](#)
- **JavaScript - The Complete Guide 2021** (Udemy, **Instructor:** Maximilian Schwarzmüller, 52.5 hrs, Sep 2021) [\[Link\]](#)
- **React - The Complete Guide** (Udemy, **Instructor:** Maximilian Schwarzmüller, 48 hrs, Sep 2021) [\[Link\]](#)
- **Blockchain A-Z** (Udemy, **Instructor:** Hadelin de Ponteves et al., 14.5 hrs, Sep 2021) [\[Link\]](#)
- **React Native - The Practical Guide** (Udemy, **Instructor:** Maximilian Schwarzmüller, 17.5 hrs, Sep 2021) [\[Link\]](#)
- **Python - The Practical Guide** (Udemy, **Instructor:** Maximilian Schwarzmüller, 17.5 hrs, Sep 2021) [\[Link\]](#)
- **Node.js - The Complete Guide** (Udemy, **Instructor:** Maximilian Schwarzmüller, 33 hrs, Sep 2021) [\[Link\]](#)
- **MongoDB - The Complete Guide** (Udemy, **Instructor:** Maximilian Schwarzmüller, 40.5 hrs, Sep 2021) [\[Link\]](#)
- **React, NodeJS, ExpressJS & MongoDB - The Mern Full Stack Guide** (Udemy, **Instructor:** Maximilian Schwarzmüller, 17.5 hrs, Sep 2021) [\[Link\]](#)
- **Kick-Start Your Career** (BYLCx, Bangladesh Youth Leadership Center, Feb 2022) [\[Link\]](#)

Projects

SecureUS | [Javascript](#), [TypeScript](#), [React.js](#), [Redux](#), [Express.js](#), [MySQL](#), [MongoDB](#) | [Site](#) | [Code](#)

- Developed a full-stack MERN web application for a women’s cybersecurity awareness research project, featuring a forum for incident sharing and a RESTful API with React frontend.
- Implemented JWT-based authentication and enabled users to book appointments with expert consultants.
- Used Celery and Redis to handle asynchronous tasks for improved performance.

Crypto Alert | [Figma](#), [Push API](#), [React](#), [React Native](#), [Express.js](#), [Javascript](#), [Git](#), [Docker](#) | [Code](#)

- Developed a cryptocurrency price alert system that notifies users when to enter or exit the market based on parameters they set.
- Implemented push notifications to alert users in real-time, allowing them to set the priority level of each notification.
- Implemented algorithms such as Stochastic RSI to predict market trends and notify users when to enter or exit the market.
- Implemented a proxy API using Express.js and MongoDB to automatically switch to a backup server if the primary server shuts down, ensuring continuous service availability.

Technical Skills

Research Skills: Semi-structured interviews, surveys, cognitive walkthroughs, think-aloud protocols, thematic analysis, user study design, and controlled studies.

Languages: JavaScript, TypeScript, Solidity, Java, Python, HTML/CSS, GraphQL

Databases: MySQL, MongoDB

Frameworks: React.js, Redux, Node.js, Express.js, Ganache, Truffle, Mocha, Chai, Spring boot, Material-UI, Tailwind

Development Tools: Git, Docker, GCP, AWS, Linux, OpenStack Swift, VS Code, PyCharm, Figma, IntelliJ, Draw.io

Libraries: Pandas, NumPy, Matplotlib

Project Management: JIRA, Slack

REFEREE

Dr. Md Sadek Ferdous

Professor, Department of CSE

BRAC University

sadek.ferdous@bracu.ac.bd

Dr. Farida Chowdhury

Professor, Department of CSE

BRAC University

farida.chowdhury@bracu.ac.bd

Dr. S M Taiabul Haque

Professor, Department of Computer Science and Engineering

BRAC University

taiabul.haque@bracu.ac.bd